

Philips Biosensing Platform

TS1611 SpO₂ Measurement and Calibration

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1 Introduction

1.1 Purpose

To provide guidance on how to perform SpO₂ calibration and measurement for Philips algorithm.

1.2 Scope

This document describes the procedure to calibrate SpO₂ performance on a PPG device with Philips algorithm to obtain an accurate SpO₂ indication. It also explains to some extent the cause of deviations that can occur due to sensor specifics.

1.3 Definitions, Acronyms and Abbreviations

Abbreviation	Description
ABG	Arterial Blood Gas. This method of blood analysis can provide, amongst others, the oxygen level carried by the red blood cells.
CPAP	Continuous Positive Airway Pressure
FiO ₂	The fraction of inspired oxygen
LED	Light Emitting Diode
PD	Photo diode
PPG	PhotoPlethysmoGraphy
PSP	Philips Biosensing Platform
SpO ₂	Peripheral oxygen saturation. This reading indicates the amount of oxygen being carried by red blood cells
DUT	Device under test

1.4 References

Abbrev.	Title	Author(s)	Version
N/A			

2 What is calibration

The SpO₂ metric in the PSP library measures the blood oxygen level in a similar way as the pulse oximeters. To measure, it makes use of the reflection of light signals on the photodiode (PD) coming from a set of GREEN, RED and InfraRed(IR) LED's. The wavelength's of both the RED and IR LED's are important in the calculation of SpO₂ while GREEN LED's are used for improving the accuracy. The recommended wavelength for RED and InfraRed are 660nm and 880nm respectively, to keep the variations minimal.

The positioning of the LED's with respect to the PD's on the wrist may influence the accuracy of the calculation. Therefore, the SpO₂ algorithm is better to perform a calibration for a specific sensor design, i.e. LED and PD positioning and chosen wavelength's, to match algorithm settings and sensor specifics.

The basic principle of estimating the SpO₂ value is to calculate the ratio between RED and IR signals. The theory can be found in literature and beyond the scope of this document.

The SpO₂ algorithm uses a parameter calibration procedure by fitting a curve to the measured reference levels. The blue line in Figure 1 is the curve with the best fit from the cloud of measurements points (grey). The ratio on the horizontal axis is an indication of the measured SpO₂ level. What happens typically after changes in the sensor or wavelength's is that the curve is not the best fit anymore (i.e. if the default curve would be used), leading to deviations in the measured SpO₂ levels. If that is the case, new calibration parameters are required corresponding to the fitted curve in blue.

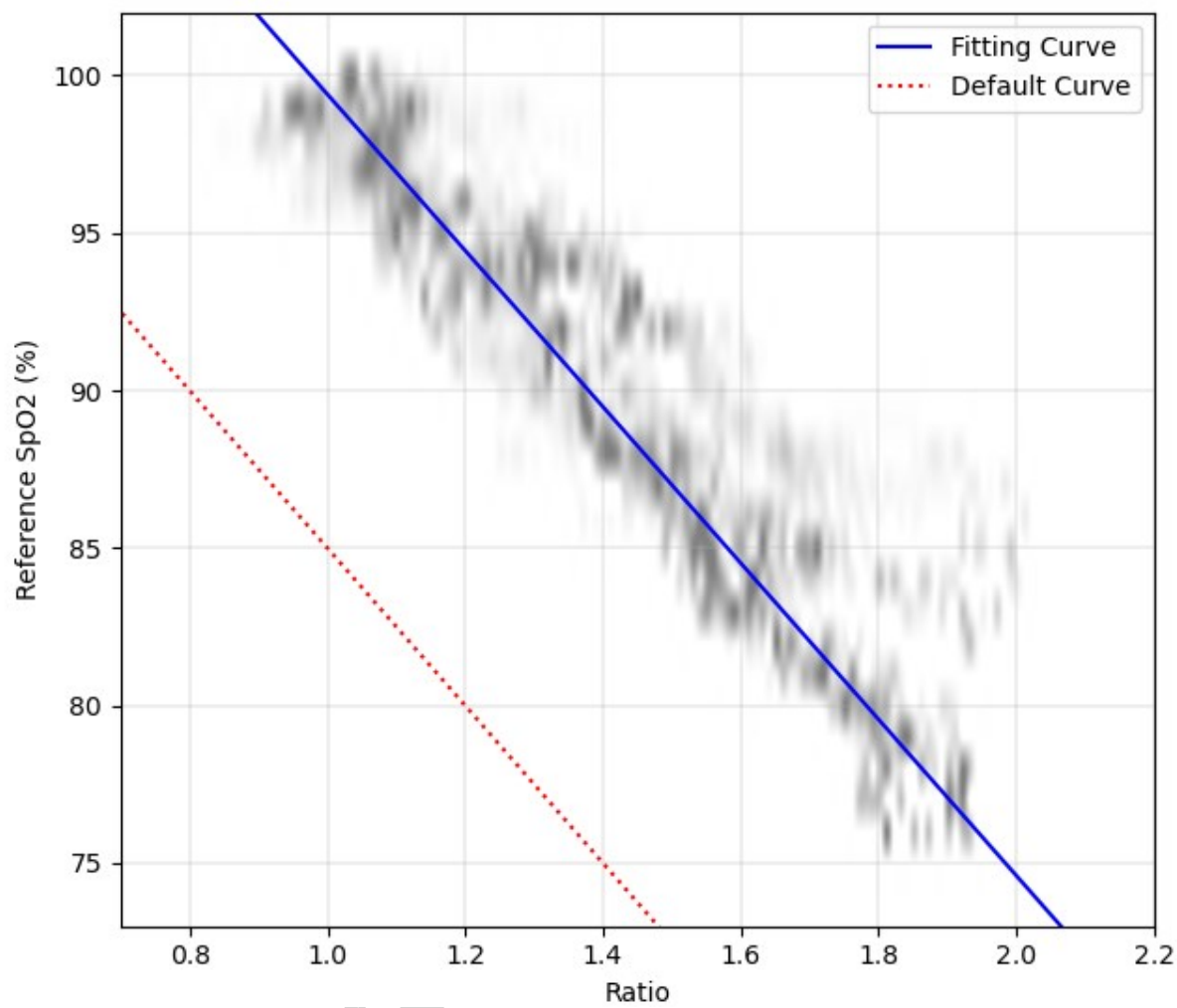


Figure 1 : Calibration Curve

3 Obtain calibration data

The SpO₂ metric can be calibrated by two methods :

1. Wellness domain – measurement targeting >95% SpO₂
2. Medical domain (FDA clearance) – Desaturation test (>70% SpO₂)

Both calibration methods need a data set with a reference device (ABG and/or pulse oximeter). For consumer products where only SpO₂ levels in the range of 95-100% (Wellness domain) are measured, the pulse oximeter is sufficient as a reference.

For products that eventually would like to obtain an FDA clearance (Medical domain), both references ABG and pulse oximeter are possible. The choice of the pulse oximeter reference device has to be the one approved by the (local) government authority. It should also provide a continuous logging function for heart rate and SpO₂ with at least 1Hz frequency.

If the reference device is the pulse oximeter that attaches to a finger, it's optional to have the reference device and the watch/band type DUT on both left and right hands. However, if the DUT belongs to the ring type, we suggest the reference pulse oximeter and ring-type DUT be on the same finger of the same hand. In such a case, for a single run of the test, there are double data points collected for 1 subject.

In both methods, during the calibration, the following data should be captured:

- Raw RED, InfraRed, GREEN (and ambient if any) PPG signals from the DUT
- Accelerometer signals from the DUT
- SpO₂ value from the reference device
- Pulse rate value from the reference device

Also, for watch/band type DUT, the participant should wear the DUT on the wrist with a strap under appropriate tightness. In addition, during measurement, subject is required to sit still. It is suggested to put the arm on the table with the device not touching the table (e.g. left hand resting on the right hand) and not moving the wrist or fingers. The hand position should not cause the device backplate lifting up on the wrist.

For ring type DUT, below are the notes for calibration preparation. Note for the moment, supporting ring form factor by PSP is experimental.

- Finger: All subjects should wear the ring on the same finger, preferring the middle finger or thumb. If data from different fingers is provided, use the file name or folder to properly distinguish.
- Wearing position: The ring rotates to a fixed position for all subjects, e.g., LEDs and PDs contacting the bottom (palm) side of the finger.
- Fitting: Snugly, but not too tight to impair blood flow. Also provide the finger size of each subject.
- Gesture for the taking measurement: The hand with the probes relaxed, without movement, and resting on a table (the hand is at about the same level as the elbow). The ring should not touch the table.

- Warm-up: After putting on the ring every time, wait about 60 seconds before starting the actual measurement to allow the finger to adjust to the pressure of the ring. Besides, the hand should be warm to ensure good blood perfusion.
- Reference device (clip type): The ring and the ref device are placed on the same finger of the same hand.

To ensure smooth execution of data measurement, it is strongly recommended to have a trial run test with only 1 or 2 subjects before the official test. Record the detailed test setup, steps. Log the data and send them to Philips technical support for sanity check.

3.1 Methods targeting Wellness domain – measurement of SpO₂ >95%

This method is intended to be carried out under normal condition without involving a laboratory.

This method requires at least 20 healthy subjects and uses a pulse oximeter as a reference, e.g. Nellcor PM10N. The inclusion and exclusion criteria of the subjects are given below.

Inclusion Criteria:

1. Male and female subjects, aged ≥ 22 and < 65 .
2. Minimum 30% of each gender.
3. 15% of subjects should be dark skin with Fitzpatrick scale V or VI.
4. The subject is in good general health with no evidence of any medical problems.
5. The subject has provided informed consent and is willing to comply with the study procedures.

Exclusion criteria:

1. The subject is obese or too thin (BMI > 35 or BMI < 18.5).
2. The subject has tattoos, skin lesions, skin lotions or nail polish at sensor sites (wrist and finger).
3. The subject is a current smoker.

The calibration should follow the steps below:

1. Subject straps device to left wrist (dorsal side, before the ulnar towards the elbow). Adjust correct tightness.
2. Arm with device resting on table with the device not touching the table (e.g. left hand resting on the right hand) and not moving the wrist or finger.
3. Attach Nellcor reference to the middle finger of the left hand (cable site on the nail).
4. Measures 3 minutes of SpO₂ data with the device and Nellcor reference (record RAW PPG & ACC data + Nellcor data (SpO₂ and heart rate))

5. Removes strap and Nellcor
6. Take a one minute break
7. Repeat the above steps 4 more times so we have 5 recordings from the same subject
8. Change the subject to the next person
9. Repeat steps above steps until we have 5 recordings of in total 20 different subjects

The data format is preferably stored with Philips FT format. The dataset could be used to calculate the calibration parameters.

3.2 Method targeting Medical domain (FDA clearance) – Desaturation test (>70% SpO₂)

In this method, the measurement is intended to be carried out in a hypoxia laboratory under medical surveillance, subjects will undergo a desaturation wearing a mask where the oxygen levels are gradually reduced to as low as reaching a 70% SpO₂ level. In these laboratories, usually both references, ABG or pulse oximeter, can be chosen (at different cost).

The description of the method below is taken from a typical procedure from a hypoxia laboratory in the US.

A typical study will include at least 20 subjects. Per FDA guidance, at least 2, or 15% of the subjects will have dark skin with Fitzpatrick scale V or VI. Equal numbers of men and women will be enrolled. Besides, the subjects are chosen with the following inclusion and exclusion criteria.

Inclusion Criteria:

1. Male and female subjects, aged ≥ 22 and < 65 .
2. The subject is in good general health with no evidence of any medical problems.
3. The subject is fluent in both written and spoken English.
4. The subject has provided informed consent and is willing to comply with the study procedures.

Exclusion criteria:

1. The subject is obese (BMI > 30).
2. The subject has tattoos, skin lesions, skin lotions or nail polish at sensor sites (wrist and finger).
3. The subject has a known history of heart disease, lung disease, kidney or liver disease.
4. Diagnosis of asthma, sleep apnea, or use of CPAP.
5. Subject has diabetes.
6. Subject has a clotting disorder.

7. The subject has a hemoglobinopathy or history of anemia, per subject report or the first blood sample, that in the opinion of the investigator, would make them unsuitable for study participation.
8. The subject has any other serious systemic illness.
9. The subject is a current smoker.
10. Any injury, deformity, or abnormality at the sensor sites that in the opinion of the investigators would interfere with the sensors working correctly.
11. The subject has a history of fainting or vasovagal response.
12. The subject has a history of sensitivity to local anesthesia.
13. The subject has a diagnosis of Raynaud's disease.
14. The subject has unacceptable collateral circulation based on exam by the investigator (Allen's test).
15. The subject is pregnant, lactating or trying to get pregnant.
16. The subject is unable or unwilling to provide informed consent or is unable or unwilling to comply with study procedures.
17. The subject has any other condition, which in the opinion of the investigators would make them unsuitable for the study.

A typical calibration of the SpO₂ metric by desaturation could be performed by gradually decreasing the oxygen level from 21% FiO₂ to 10% FiO₂. During the decrease, there should be six plateaus (e.g. SpO₂ of 94%, 90%, 85%, 80%, 75%, 70%). The plateau can be indicated by checking the SpO₂ value of the reference device. In case there are more than one subject in the room, the first person reaching the required SpO₂ level determines the plateau. Each plateau should be held for at least one minute, because of stabilization of the PPG signal. The duration of the calibration with desaturation is depending on the possibility to desaturate the room and stabilize the oxygen level in the room (could take 1 hour). The subject should undergo the situation that the FiO₂ is desaturating gradually instead of dramatic change. It is more realistic and also safer. In other words, it should not happen that a subject is walking into a desaturated room with 10% FiO₂ from the open air. The precise target levels of saturation can be adjusted to suit the sponsor, but typical testing is done to satisfy ISO and FDA standards for testing, which is SpO₂ of 70% to 100%.

The calibration steps should be worked out together with the laboratory. The dataset gathered could be used to calculate the calibration parameters.

4 The practice for PSP library

For the development of the SpO₂ algorithm in the PSP library, calibration data has been recorded for the sensor geometry as depicted in Figure 2. The recordings were done in the spring of 2021 and in the hypoxia laboratory in June 2021. The picture shows the layout of the sensor. (after mounting in a watch, the clock positions are seen from the top of the watch).

- RED LED's (660nm) positions 12 and 6 o'clock and in the middle
- IR LED's (880nm) positions 12 and 6 o'clock
- Green LED's (560nm)

Note: the calculation of the SpO₂ data, uses the photodiodes in the clock position 12, 6 in combination with the RED and InfraRed LED's in the 6,12 position.

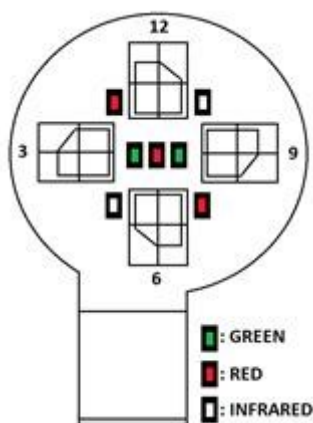


Figure 2

The data sets for the calibration of the algorithm (raw PPG and ACC data) are recorded in a hypoxia lab and use the Nellcor finger clip as the SpO₂ reference device. The typical desaturation plateaus as described in the previous section are used.

Calibration parameters can be derived from this dataset particular to this specific geometry and wavelength.

Appendix A: Commercially accessible hypoxia labs

UCSF: <https://hypoxialab.com/contact>.

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